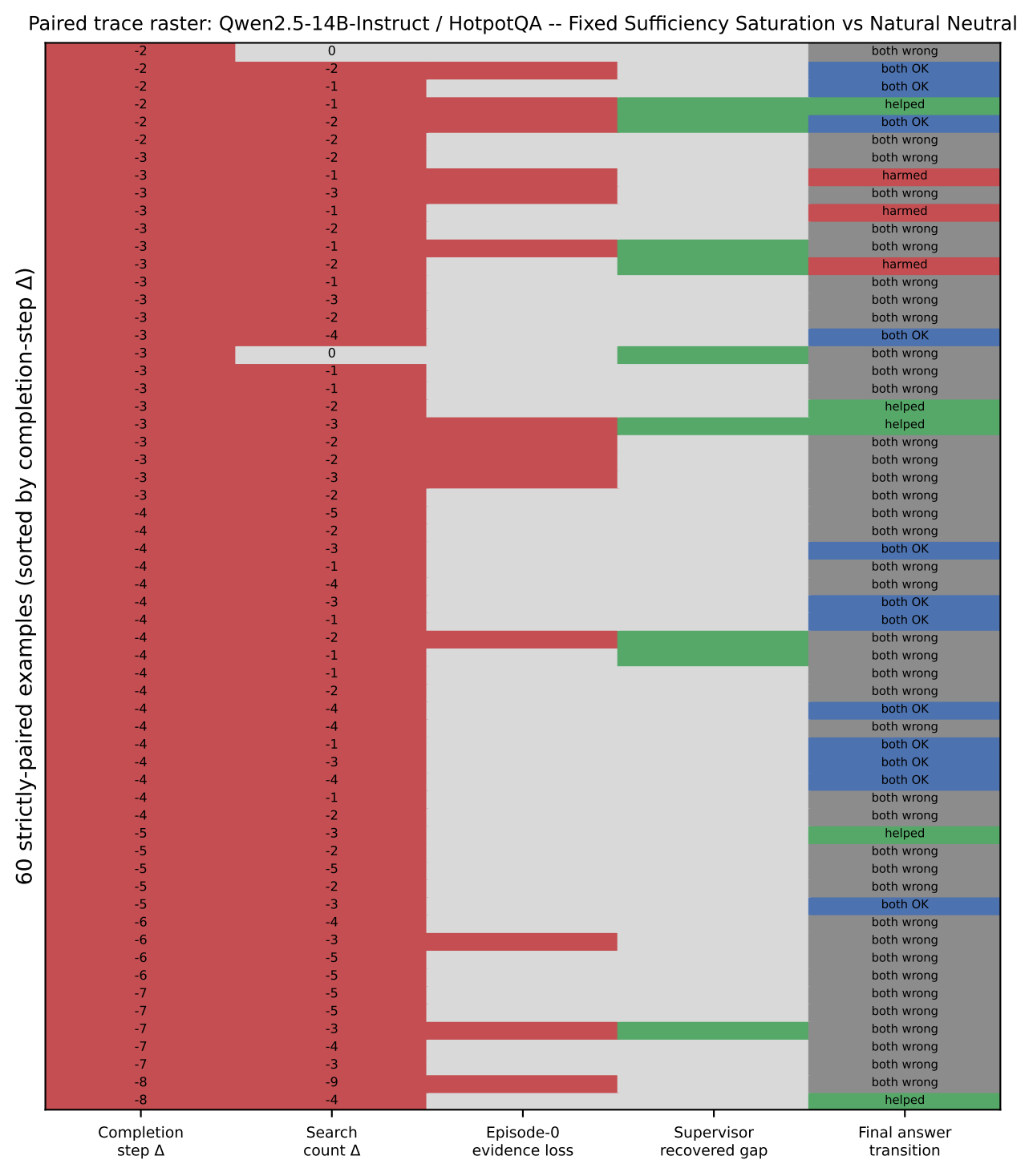
Paired trace raster: Qwen2.5-14B-Instruct / HotpotQA -- Fixed Sufficiency Saturation vs Natural Neutral
60 strictly-paired examples (sorted by completion-step Δ)
-2 0 both wrong
-2 -2 both OK
-2 -1 both OK
-2 -1 helped
-2 -2 both OK
-2 -2 both wrong
-3 -2 both wrong
-3 -1 harmed
-3 -3 both wrong
-3 -1 harmed
-3 -2 both wrong
-3 -1 both wrong
-3 -2 harmed
-3 -1 both wrong
-3 -3 both wrong
-3 -2 both wrong
-3 -4 both OK
-3 0 both wrong
-3 -1 both wrong
-3 -1 both wrong
-3 -2 helped
-3 -3 helped
-3 -2 both wrong
-3 -2 both wrong
-3 -3 both wrong
-3 -2 both wrong
-4 -5 both wrong
-4 -2 both wrong
-4 -3 both OK
-4 -1 both wrong
-4 -4 both wrong
-4 -3 both OK
-4 -1 both OK
-4 -2 both wrong
-4 -1 both wrong
-4 -1 both wrong
-4 -2 both wrong
-4 -4 both OK
-4 -4 both wrong
-4 -1 both OK
-4 -3 both OK
-4 -4 both OK
-4 -1 both wrong
-4 -2 both wrong
-5 -3 helped
-5 -2 both wrong
-5 -5 both wrong
-5 -2 both wrong
-5 -3 both OK
-6 -4 both wrong
-6 -3 both wrong
-6 -5 both wrong
-6 -5 both wrong
-7 -5 both wrong
-7 -5 both wrong
-7 -3 both wrong
-7 -4 both wrong
-7 -3 both wrong
-8 -9 both wrong
-8 -4 helped
Completion step Δ
Search count Δ
Episode-0 evidence loss
Supervisor recovered gap
Final answer transition